Case 3: Modeling Stock Prices, Returns and Volatility

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In this case you will explore financial time series, from modeling stock prices and returns using ARIMA models to modeling their volatility using GARCH and HAR models. You will use daily stock price data of your own choosing to construct price, return and volatility time series, and you will extend your analysis to a multivariate spillover setting across several stocks. You will learn (i) how to model price and return series using ARIMA models, (ii) what GARCH models are, how to estimate them and how they are useful for modeling conditional volatility in finance, (iii) how to construct realized variance and range measures, (iv) what HAR models are, how they relate to and differ from GARCH models, and how to estimate them, (v) how to extend your univariate analysis to a multivariate one to model spillovers across stocks.

General Guidelines. The case study is deliberately formulated in a rather "open" way as you will have to "learn by doing" how to solve the tasks at hand. This document provides you with initial pointers for additional reading material (academic papers, software papers, ...) that may be relevant. You may resort to any other auxiliary material that helps you to solve this case. When presenting your case study in class, make sure to

- Prepare compact "lecture material" (for instance slides) when you need to introduce new time series concepts, models, estimators, ... to your fellow students.
- Prepare a careful analysis in `R` using readable scripts and intuitive figures, ... such that you can walk your fellow students through your software implementation. While you can make use of any auxiliary material, you should be able to explain all the code you generated in `R` (motivate use of certain packages and functions, know how the functions work, what arguments in the functions mean, ...)! You should also spend sufficient time on the interpretation of the output and the results. Make sure that you `R` scripts are in a format that allows for easy replication.
- Be transparent about, and take full responsibility for, any possible use of external tools, including generative AI. You may use such tools to support your work, but you must be able to explain, motivate and defend every (methodological) choice, every line of `R` code and every result live during the in-class presentations and Q&A. Answers that reveal a lack of genuine understanding of the underlying methodology, regardless of how they were produced, will be graded accordingly. Each team will also upload a reflection document on GenAI usage, see Canvas for more details.

Data. The data come from Yahoo Finance, a widely-used, publicly available source of historical stock price data, accessible at `https://finance.yahoo.com` or programmatically in `R` via packages such as `quantmod` (Ryan and Ulrich, 2025). Choose a stock of your own choosing that is listed on Yahoo Finance and for which sufficient historical daily price data are available. Motivate your choice. Collect daily price data for your chosen stock. Specifically, the following variables are typically available

- `Date`: the trading date
- `Open`, `High`, `Low`, `Close`: the opening, highest, lowest and closing price on the trading day
- `Adjusted Close`: the closing price adjusted for dividends and stock splits
- `Volume`: the number of shares traded on the trading day.

Make sure that your time series is sufficiently long: several of the models you will estimate later in this case require a substantial number of observations to be estimated reliably and to cover at least one period of financial crisis. Motivate your choice of sample length.

Data Exploration. Start by importing your data into `RStudio` and visually exploring the price data. You should familiarize your fellow students with the data and its typical features. Think about discussing trends, stationarity, volatility clustering, structural breaks (e.g. financial crises), and other typical features you notice in the price series.

Unit Root Tests for the Price Series. To determine whether your stock price series is stationary you can use unit root tests. Perform an appropriate unit root test, using the Pantula principle, on the stock price series to determine its order of integration. Which unit root test do you use? What do you conclude?

ARIMA Models for the Price Series. Model your price time series using ARIMA models, that you know from the lecture. Explore two routes: (i) specify yourself appropriate ARIMA models that seems suited for the price data by relying on tools such as the correlogram. (ii) resort to automatic searches across the whole class of ARIMA models to select an appropriate model in a data-driven way, see for instance Hyndman and Khandakar (2008). In both situations: motivate how you made your choices, make sure to explain to your fellow students how the approaches work, and present your estimated models! Also, be precise on which `R` packages and functions you use to estimate the ARIMA models, motivate your choice, and introduce your fellow students to your analysis in `R`.

From Prices to Returns. Prices are typically not the object of interest in financial time series modeling: it is more common to work with *returns*. Construct the return series from your price series (e.g. log returns or simple returns; motivate your choice). Discuss basic properties of your return series: is it stationary? Perform an appropriate unit root test and compare your conclusion to the one you obtained for the price series. Model your return time series using ARIMA models. What do you find? How does the fit of your ARIMA model for the returns compare to the fit for the prices, and how do you explain this?

Modeling Volatility: GARCH Models. Return series are hard to model and forecast using the mean-based ARIMA framework: financial returns are typically close to unpredictable in their conditional mean. Nonetheless, they often display pronounced volatility clustering (periods of high volatility followed by more high volatility, and vice versa). This suggests that, rather than modeling the conditional mean of the returns, we may want to shift focus to modeling their conditional variance, i.e. their volatility, instead.

Explore the class of Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models to this end, see Bollerslev (1986) or Tsay (2010) for a textbook introduction. First, introduce your fellow students to (G)ARCH models: which stylized facts of financial returns do they capture, and how do they differ conceptually from the ARIMA models you used above? Second, find an appropriate `R` package and estimate appropriate (G)ARCH models for your return series. Present and interpret your estimated models.

HAR Models for Realized Volatility. GARCH models are not the only way to model conditional volatility. A recent and attractive alternative is the class of Heterogeneous Autoregressive (HAR) models for realized volatility, see Corsi (2009), building on the realized volatility literature of Andersen et al. (2003).

First, explain to your fellow students the concept of realized variance, and explain how HAR models differ from GARCH models: how do the two classes of models treat volatility, what data do they require, and what does "heterogeneous" refer to in HAR? Second, if you would like to estimate a HAR model for your stock, what data do you require?

Bridging the Data Gap: Realized Range. Standard HAR models are built using data that you likely do not have access to for your chosen stock. We will proceed in two ways to bridge this gap. First, consider computing the daily realized range instead of the daily realized variance, see Christensen et al. (2009).

What is the realized range, and how does it relate to the realized variance? What additional data do you need to collect to compute it for your stock (think back to the variables you already collected in the data section)? Compute the realized range for your data, and estimate a HAR-style model for your realized range series. Discuss and, informally, compare your results to those of your GARCH model.

Bridging the Data Gap: Monthly Realized Variance from Daily Data. Second, instead of using intraday data to compute a daily realized variance, we can use daily data to compute a lower-frequency, monthly realized variance, in an analogous fashion to how the daily realized variance is computed from intraday returns, see for instance Corsi (2009).

Explain to your fellow students how you would construct a monthly realized variance measure from your daily return data. Compute this monthly realized variance for your data, and estimate a HAR-style model for it. Discuss your results.

As of here, collect additional stock price series and extend your favorite univariate model to a multivariate setting.

A Multivariate Extension: Modeling Spillovers. So far, you have considered only a single stock price series, and explored several univariate models for its price, return and volatility dynamics. Choose your favorite univariate model, for your favorite time series (price, return, or volatility) among those you have used so far, and explore a multivariate extension of it. Multivariate models allow you to explore spillovers between different stock series, see for instance Diebold and Yılmaz (2015), while Hecq et al. (2023), or Diebold and Yilmaz (2012) may serve as sources of inspiration for which multivariate models to use.

Collect additional stock series (for instance further companies in the same sector, or the broader market). The goal is to model spillovers between your series. Which multivariate model do you consider, and why? Explain the model to your fellow students. Find a suitable `R` implementation, and discuss your results.

Subsample Analysis: Crisis versus Non-crisis Periods. First, perform the spillover analysis of the previous section for your whole sample. However, your sample likely spans one or more periods of financial crisis. Perform a subsample analysis: split your sample into a crisis and non-crisis period, motivating your choice of split. How do the results of your spillover analysis differ between the crisis and the non-crisis period?

Final Recommendations. Conclude with your lessons learned, tips & tricks and final recommendations for this case.

References

Andersen, T. G.; Bollerslev, T.; Diebold, F. X. and Labys, P. (2003), "Modeling and forecasting realized volatility," *Econometrica*, 71, 579–625.

Bollerslev, T. (1986), "Generalized autoregressive conditional heteroskedasticity," *Journal of Econometrics*, 31, 307–327.

Christensen, K.; Podolskij, M. and Vetter, M. (2009), "Bias-correcting the realized range-based variance in the presence of market microstructure noise," *Finance and Stochastics*, 13, 239–268.

Corsi, F. (2009), "A simple approximate long-memory model of realized volatility," *Journal of Financial Econometrics*, 7, 174–196.

Diebold, F. X. and Yilmaz, K. (2012), "Better to give than to receive: Predictive directional measurement of volatility spillovers," *International Journal of Forecasting*, 28, 57–66.

Diebold, F. X. and Yılmaz, K. (2015), *Financial and macroeconomic connectedness: A network approach to measurement and monitoring*, vol. 121, Oxford University Press Oxford.

Hecq, A.; Margaritella, L. and Smeekes, S. (2023), "Granger causality testing in high-dimensional VARs: A post-double-selection procedure," *Journal of Financial Econometrics*, 21, 915–958.

Hyndman, R. J. and Khandakar, Y. (2008), "Automatic time series forecasting: The forecast package for R," *Journal of Statistical Software*, 27, 1–22.

Ryan, J. A. and Ulrich, J. M. (2025), *quantmod: Quantitative Financial Modelling Framework*, r package version 0.4.28.

Tsay, R. S. (2010), *Analysis of financial time series*, vol. 3, wiley New York.